

Information Technologies for Industrial Engineers

JavaScript

- JavaScript is the Programming Language for the Web
- Can be used on server-side
 - `Node.js`
- [Stack Overflow Survey 2025](#)

TypeScript

- TypeScript is a syntactic superset of JavaScript which adds static typing.
- It shares the same base syntax as JavaScript, but adds something to it.
- *Much beter than JavaScript, trust me.*

Initializing a variable

```
let myName = "Me";  
const yourName = "You";
```

Variable types

```
const numberType = 1;  
const stringType = "This is a string";  
const booleanType = true;  
const undefinedType = undefined;  
const nullType = null;  
const arrayType = [1, 2, 3];  
const objectType = { key1: 1, key2: "two" };
```

To check type

```
console.log(typeof numberType);
```

Function

```
// Function
function hello1(name: string) {
  console.log("Hello " + name);
}

// Arrow function
const hello2 = (name: string) => {
  console.log("Hello " + name);
};

// Calling
hello1("Ari");
hello2("Tom");
```

if - else if - else

```
let choice = "A"; // 'A', 'B', 'C'

if (choice === "A") {
  console.log("You chose A.");
} else if (choice === "B") {
  console.log("You chose B.");
} else if (choice === "C") {
  console.log("You chose C.");
} else {
  console.log("You did not choose A, B, or C.");
}
```

Looping code

- Print 10 random numbers

```
for (let i = 0; i < 10; i++) {  
  console.log(Math.random()); // 10 randoms numbers  
}
```

Looping through an array

```
const cats = ["Leopard", "Serval", "Jaguar", "Tiger", "Caracal", "Lion"];

for (const cat of cats) {
  console.log(cat);
}
```